

Kitchen Safety Heat and Gas Monitoring System

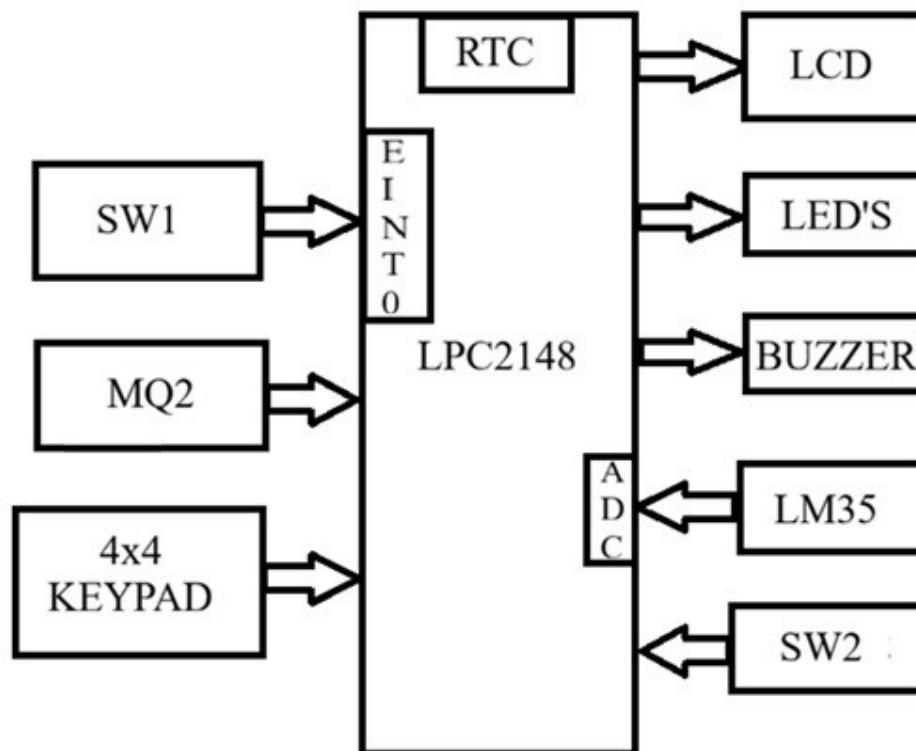
AIM:

To design and implement a smart safety system using the LPC2148 microcontroller that continuously monitors kitchen temperature and gas levels, detects hazardous conditions, records threshold-crossing events with RTC timestamps, periodically displays the most recent safety event, and provides timely alerts to prevent accidents.

Objectives:

- **Temperature Monitoring:** Continuously measure kitchen temperature using the LM35 sensor and detect abnormal heat levels.
- **Gas Leakage Detection:** Sense the presence of combustible gases (LPG, methane, smoke) using the MQ2 sensor.
- **Alert Mechanism:** Trigger alarms (buzzer, LED, or display) when unsafe conditions are detected, ensuring immediate user awareness.
- **Safety Enhancement:** Provide a reliable, low-cost embedded solution to enhance kitchen safety and prevent fire or gas-related accidents.

BLOCK DIAGRAM:



REQUIREMENTS:

HARDWARE REQUIREMENTS:

- LPC2148
- 16X2 LCD
- 4X4 MATRIX KEYPAD
- SWITCH
- LED'S
- MQ2
- LM35
- BUZZER
- USB-UART CONVERTER / DB-9 CABLE

SOFTWARE REQUIREMENTS:

- EMBEDDED-C PROGRAMMING
- FLASH MAGIC

PROJECT WORKFLOW:

The system operates in a **continuous monitoring loop**, where the **LPC2148 microcontroller** samples the current temperature from the **LM35 sensor** and gas concentration from the **MQ2 sensor** using its on-chip ADC. The measured values are displayed on a **16×2 LCD**, along with real-time information (time, date, month, and year) provided by the on-chip **RTC module**.

Whenever the **temperature exceeds the predefined threshold** or **gas leakage is detected**, the system immediately activates the **buzzer** to provide an alert indication and switch on exhauster fan by using DC motor (Instead of DC motor use LED'S). The buzzer can be silenced by pressing **Switch2**, which resets the alert condition.

In addition to real-time monitoring, the controller maintains a record of the most recent safety event. Whenever the temperature or gas sensor value crosses its configured set point from a safe condition to an unsafe condition, the system captures the current RTC timestamp, sensor name, and measured value. During normal operation, the LCD

continuously displays the current time and live sensor readings. At every 10-second interval, the LCD temporarily displays the most recently recorded event for two or three seconds, including which sensor crossed its set point, the recorded value, and the time of occurrence. The display then automatically returns to the normal monitoring screen. The stored event information is updated only when a new threshold-crossing event occurs.

Additionally, when the user presses **Switch1**, an external interrupt is triggered, and the application enters **EDIT MODE**. In this mode, the user can modify system parameters such as threshold values or RTC settings, ensuring flexibility and adaptability of the system.

Basic Level Security Inclusion

Before allowing any modification, the system asks for a basic password.

Security Check Before Editing

Once the switch press is detected:

- Prompt user to enter password through keypad
- Compare entered password with stored password
- If password is correct:
 - allow editing of parameters
- If password is wrong:
 - deny access
 - increment wrong-attempt counter
 - show “Access Denied” on LCD
 - optionally blink buzzer/LED

Wrong Attempt Handling

- If wrong password is entered 3 times:
 - lock edit mode for a short duration or until reset

- show “System Locked”
- optional alert indication through buzzer

This gives the project a simple but attractive security layer.

Editable Parameters in Secure Edit Mode

After successful password validation, allow user to edit:

- Current RTC time, day and date
- Threshold value
- Password change option
- The keypad can be used to navigate through a menu.

Return to Normal Mode

After editing is completed:

- save updated parameters
- exit edit mode and return to normal monitoring operation

The system then resumes:

- RTC display
- Temperature and Gas monitoring
- Buzzer indication

***** ALL THE BEST *****

Note:

The application code must be developed following industry-standard embedded programming practices, with proper structure, naming conventions, modular design, and complete test case validation support. Any code not adhering to these standards will not be accepted.